

From Cash to Cashless: UPI's Impact on Spending Behavior among Indian Users

Harshal Dev*
harshal19306@iiitd.ac.in
IIIT Delhi
New Delhi, India

Raj Gupta*
raj21410@iiitd.ac.in
IIIT Delhi
New Delhi, India

Dhruv Kumar
dhruv.kumar@iiitd.ac.in
IIIT Delhi
New Delhi, India

ABSTRACT

The emergence of digital payment systems has transformed how individuals conduct financial transactions, offering convenience, security, and efficiency. One groundbreaking innovation making waves in the Indian financial landscape is the Unified Payments Interface (UPI). Existing work has explored how digital payments benefit a country's economy and GDP. However, our study explores how the introduction of UPI has influenced spending behavior among Indian users on an "individual" level. We gathered 235 valid survey responses encompassing diverse demographics and interviewed 20 survey respondents. Approximately 75% of the survey respondents reported increased spending due to UPI, with only 7% indicating reduced spending. Significantly, 91.5% of the respondents reported satisfaction with their UPI usage. Also, 95.2% of the survey respondents found making payments via UPI convenient. Our research also provides suggestions for UPI applications and various stakeholders to enhance digital payment systems, enabling users to make informed decisions and fostering responsible financial management.

CCS CONCEPTS

• **Human-centered computing** → *Empirical studies in HCI*.

KEYWORDS

Digital Payments, Cashless Transactions, Unified Payments Interface (UPI), User Study

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1 INTRODUCTION

An E-payment is generally understood as an economic situation where the buying and selling of goods are handled without physical cash, often facilitated through E-payment or checks [12]. Digital payments have become integral to modern financial landscapes,

offering swift, secure, and convenient electronic transactions that transcend traditional methods. This global shift towards digital transactions has been accelerated by the widespread adoption of internet and mobile technologies, reshaping financial practices across the world [48]. In 2022 alone, the volume of real-time digital transactions reached an unprecedented 195 million [42].

India, in particular, has emerged as a global leader in digital transactions, accounting for 46 % of the total global digital transactions in 2022 [42]. Unified Payments Interface (UPI), launched in 2016 by the National Payments Corporation of India (NPCI), has redefined financial transactions by enabling instant, real-time inter-bank transfers [59]. As of August 2023, UPI has facilitated a staggering 10 billion transactions, totaling Rs 18,22,949.42 Crore (189 billion USD) [45]. In India, 84 % of all digital transactions in 2022, amounting to 1.79 trillion dollars, were channeled through UPI [46]. UPI's rapid proliferation and high penetration in the Indian payment ecosystem underscore its transformative impact [59].

Despite UPI's monumental success, a critical research gap exists regarding its influence on users' spending habits. Our study addresses this crucial gap in existing research on UPI, specifically focusing on the impact of UPI on users' spending habits. While previous studies and works [63] acknowledge UPI's success in reshaping digital payments, they fall short of providing detailed insights into how this technology influences "individual" spending behaviors. The existing literature often emphasizes macro-level implications and adoption rates. Multiple questions about changes in expenditure preferences, budgeting strategies, investment choices, long-term financial goals and overall financial behaviors remain unexplored, hindering a holistic understanding of UPI adoption's intended and unintended consequences on users' daily financial practices. This study seeks to fill these critical gaps by providing in-depth insights into the micro-level effects of UPI on spending behaviors. Moreover, there's a notable absence of research delving into potential variations in UPI's impact across different demographic groups like age, profession, and socio-economic status, etc.

We explore users' perceptions of security and how trust in UPI influences expenditure patterns, unraveling the intricate dynamics between trust, technology adoption, and financial decision-making. Additionally, we investigate the relationship between users' financial literacy levels and UPI utilization, shedding light on how understanding financial principles shapes spending choices. Our research also analyzes the existing budgeting applications, employing a competitive lens to understand its impact on users' awareness of spending patterns and financial discipline.

The significance of this study extends beyond the realm of finance. In Human-Computer Interaction (HCI), understanding how

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users interact with financial technologies is essential for designing user-friendly and effective interfaces. This study contributes valuable insights to HCI researchers, financial institutions, and policymakers, guiding the development of digital payment systems that align with users' financial needs and behaviors.

In addressing the above-mentioned gaps, this research employs a mixed-methods approach, encompassing qualitative and quantitative assessments. We gathered 276 responses covering diverse demographics and backgrounds using a survey form, out of which 235 were valid responses. Additionally, we also conducted semi-structured interviews with 20 survey respondents. This study aims to address the following research questions (**RQs**):

- (1) **RQ1**: How does the introduction of UPI influence Indian users' spending habits, and what are the observed changes: increased, decreased, or unchanged spending patterns?
- (2) **RQ2**: How do users perceive the convenience and efficiency of UPI transactions compared to traditional cash transactions, and to what extent do these perceptions shape their spending behavior?
- (3) **RQ3**: To what extent does UPI adoption vary across demographic groups, such as age and profession, and how do these variations influence users' financial practices, including spending habits and long term financial goals?

Among the participants we surveyed, 74.2% of the survey participants confirmed increased spending due to UPI, with only 7% confirming reduced spending. Significantly, 91.5% of the survey respondents reported satisfaction with their UPI usage and 95.2% of the survey respondents found making payments via UPI convenient. Our research also provides suggestions for UPI applications and various stakeholders to enhance digital payment systems, enabling users to make informed decisions and fostering responsible financial management. To the best of our knowledge, this is the first user study of UPI's impact on individual spending behaviour in India.

2 RELATED WORK

Mobile payment systems are progressively influencing people's socio-economic lives in South Asia [49–51, 54, 57, 65]. India topped the list of the five countries, namely Brazil, China, Thailand, and South Korea, for digital payments by the end of 2022 with 89.5 million transactions [42]. The sheer nature of mobile payments and their ubiquity, along with convenience, helps include individuals, eventually benefiting the country's economy [16, 51].

Historically, India has been characterized by high cash usage, with a substantial portion of the workforce engaged in the informal sector relying on cash transactions [53, 60]. The transformation from a cash-driven economy to a significant shift towards digital payments is also driven by various significant events like government's "Digital India" vision, Demonetization [27, 55, 61] and COVID-19 pandemic. According to the World Bank, over 80 million Indian adults made their first digital merchant payment after the pandemic's onset [37]. Studies have indicated the positive impact of electronic transactions on trade and economic development in India [2, 9]. The benefits include transparency, accountability, and reduced cash-related fraud, contributing to overall economic growth [62]. The surge of digital wallets and UPI platforms, including BHIM UPI [22], Paytm [7], PhonePe [21], and Google Pay

[11], has been instrumental in reshaping India's financial landscape. Internet penetration in the country rose from 14% in 2014 to over 46% in 2021, driven by affordable data tariffs introduced by regional mobile operators in 2016. This contributed to India becoming the second-largest global user base of the internet by 2018 [31, 32].

Despite progress, challenges remain, as only 35% of Indian adults use bank accounts for digital payments [35]. The low percentage (approx 5% of the Indian population) of income tax contributors emphasizes the potential role of digital payments in increasing tax collection, as highlighted by previous research [4]. The extensive penetration of smartphones in India, with 1.2 billion active mobile phone users, underscores the technological underpinnings of this digital transition [38]. In 2016, a study by Goel [23] found that technological advancements have played a significant role in empowering and introducing innovations in E-payment systems.

Research in the field has extensively explored the impact of mobile payment systems on economies and societies, focusing on their transformative effects. Notably, people from different countries and different strata of society manage finances differently [52, 56, 66]. This motivated us to understand the possible changes in their spending due to UPI and how Indian users keep track of their finances. This study addresses the micro-level effects of UPI on spending behaviors, filling gaps in existing literature focused on macro-level implications. It explores changes in expenditure preferences, budgeting strategies, and long-term financial goals. The research aims to comprehensively understand the consequences of UPI adoption on users' spending behavior while highlighting user concerns and preferences that influence the widespread acceptance of mobile payment systems in the region. The study also aims to provide insights beyond transaction numbers and uncover the factors that encourage users to trust the UPI technology, considering the backdrop of rising scams [36, 41].

Evidence suggests that individuals often under-save, fail to invest wisely, and are frequently indebted [15]. Several studies indicate that a significant fraction of individuals has a low level of financial knowledge, associated with sub-optimal financial outcomes in various areas such as retirement planning [5, 8, 13, 58]. Increasing financial literacy improves financial decision-making, leading governments worldwide to recognize it as a critical life skill, launching financial education initiatives to help young people acquire this skill [19, 34]. Multiple studies have highlighted the importance of financial inclusion and literacy, with one study pointing out how roadblocks to digital or financial literacy impede digital financial inclusion [64]. While cashless transactions promote technology literacy [30], there is a visible gap in understanding how or if it promotes financial literacy. In contrast to the existing studies, our study aims to explore how financial literacy impacts spending habits, especially with the onset of cashless transactions (UPI) in the context of India.

3 METHODOLOGY

3.1 RESEARCH DESIGN

We employed a mixed methods [1] approach to investigate the impact of UPI on individual spending behaviors and financial practices. We utilized two primary data collection techniques: surveys and interviews. We used a random sampling method for the survey

component, targeting individuals aged 18 and above. Our survey was conducted through Google Forms to ensure easy accessibility and respondent anonymity (by not asking for their personal information like name, phone number, etc). It garnered 276 responses from a diverse participant pool spanning various age groups and occupations. To ensure the quality, consistency and relevance of the responses, we implemented stringent criteria for validity. Frivolous responses containing irrelevant information were excluded, for instance, in the question regarding the initiation of UPI usage, any response indicating usage before 2016 was deemed invalid, considering that UPI was publicly launched that year. This validation process resulted in 235 valid responses. We conducted in-depth semi-structured interviews by randomly selecting interviewees from our survey respondents who expressed an interest in participating. We conducted 20 (13 male & 7 female) interviews and they provided verbal consent for the same. Survey and interview questions are presented in Appendix A and Appendix B respectively.

3.2 ETHICAL CONSIDERATIONS

In conducting this research, we implemented various ethical considerations to uphold transparency and safeguard the privacy and well-being of participants. Our study's research materials and protocols underwent thorough review and approval by our university's Institutional Review Board (IRB). Before participating in interviews, all candidates received information about the study's purpose, the voluntary nature of their involvement, and the assurance of anonymity and confidentiality. We also acknowledge the significance of capturing diverse perspectives and experiences of individuals using UPI and its influence on their spending behavior. To ensure inclusivity, we actively sought participants from various backgrounds and genders. Verbal consent was obtained from participants for recording the interviews.

3.3 LIMITATIONS

One notable constraint of this study is the geographic scope of our data collection efforts. The research exclusively focused on respondents from urban areas in India, where technology adoption and usage, including electronic payment methods like UPI, tend to be higher and well integrated into daily life. As a result, the findings may not encompass the diverse spending habits. They may not accurately represent the impact of UPI on the spending habits of those living in rural areas in India.

4 EVALUATIONS

4.1 QUANTITATIVE ANALYSIS OF SURVEY RESPONSES

Our survey received 235 valid responses from a diverse range of respondents from various demographics and backgrounds. Among the respondents, 51.6% identified as students, 42% as working professionals, and 4.7% as business owners. Regarding gender distribution, the data shows that 77% of respondents were male, 19.7% were female, and 3.3% preferred not to disclose their gender. Regarding the adoption timeline of UPI (figure 1), we can conclude that the diverse range of adoption dates reflects users' varying entry points

into the UPI ecosystem, often aligning with significant events like demonetization and pandemic-related lockdowns.

To understand the impact of UPI adoption, 74.2% of the survey respondents admitted that UPI had led to increased spending, while only 7% said that it had decreased spending. Additionally, 18.8% of the survey respondents felt no changes in their spending habits after adopting UPI. Regarding satisfaction levels (figure 3), 55% of the survey respondents indicated they were highly satisfied with their overall UPI experience. Additionally, 36.5% reported satisfaction, and 7.7% remained neutral in their assessment. Therefore we can conclude that 91.5% (55% + 36.5%) of the respondents are satisfied with their overall experience using UPI for payments. When asked to rate their convenience with UPI for making payments (figure 7), the survey results indicate that 69.50% of respondents rated "Highly convenient", 25.7% rated "Convenient, and 3.7% were "Neutral" on the same. Therefore we can conclude that 95.2% (69.5% + 25.7%) of the respondents find using UPI for payments convenient. Notably, only 2-3 respondents reported dissatisfaction with UPI as a payment method, indicating that most were satisfied with their experience.

Respondents reported a varied daily spending range using UPI, from Rs 50 (0.60 USD) to 1 lakh (1,202.66 USD). On average, 15.7% mentioned a daily spending of around Rs 200 (2.41 USD). Regarding the frequency of UPI usage for payments (figure 2), a significant 81.1% use it daily, 15.6% use it weekly, 1.1% use it monthly, 1.90% use it occasionally, and 0.30% use it rarely. This data highlights the widespread and frequent adoption of UPI among surveyed individuals.

When asked about overspending due to UPI usage (figure 5), 59.8% of respondents acknowledged experiencing such situations, while 39.8% believed they had not. Additionally, respondents had diverse opinions when asked about the impact of using UPI on their budget adherence (figure 4); approximately 23% strongly agreed, 23.3% agreed, 26.7% remained neutral, 15.6% disagreed, and 11.5% reported strongly disagree, indicating that UPI has had an impact on their ability to stick to their budget and control overspending.

Respondents employed various methods to manage their expenses, like checking their transaction history, Splitwise [10] SBI YONO [28], Samsung Notes [26], using excel sheets and dedicated finance apps like MyMoney [39], Honeybook [17], Monefy [25], Payhawk [29] and Money Manager [33]. This observation highlights diverse financial management approaches among UPI users, showcasing varied responses and opinions. A significant proportion acknowledges increased spending along with high overall satisfaction with UPI and its convenience. Our comprehensive analysis of various financial tracking and budget management apps in the Indian market is summarized in Table 1 in Section 5.

4.2 QUALITATIVE ANALYSIS OF INTERVIEWS

4.2.1 EXPLORING UPI USAGE FOR TRANSACTIONS: INSIGHTS AND PERSPECTIVES.

Respondents appreciated UPI's convenience and cashless transactions, obviating the requirement to carry tangible currency. They consistently praised the system's remarkable speed and efficiency in handling small transactions.

According to a State Bank of India (SBI) report reveals that with every Rs 1 (0.012 USD) increase in UPI transactions, there is an 18

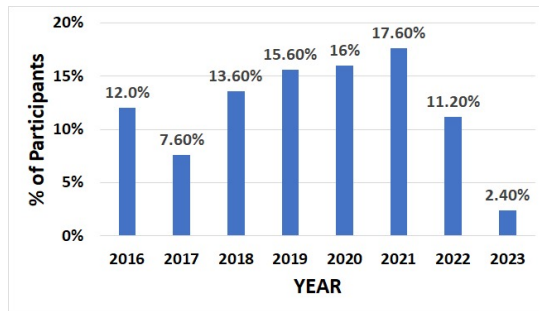


Figure 1: When did users start using UPI?

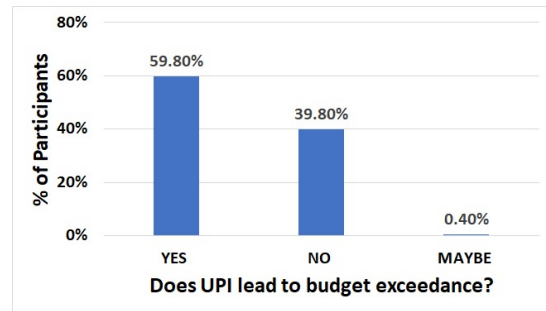


Figure 5: Budget exceedance due to UPI

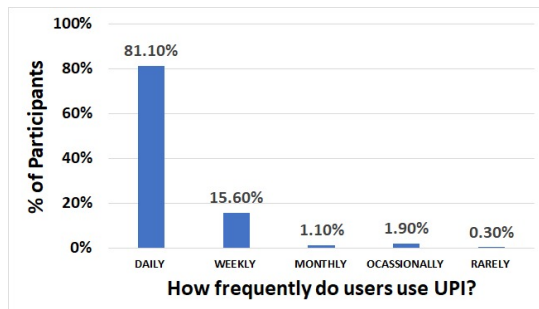


Figure 2: How often do users use UPI?

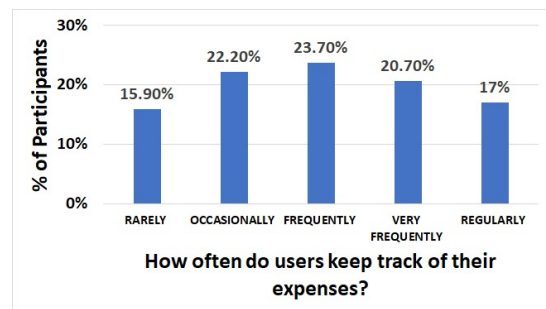


Figure 6: How often do people keep track of transaction history?

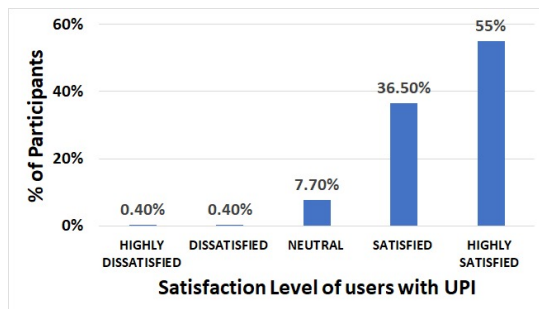


Figure 3: Satisfaction Level with UPI

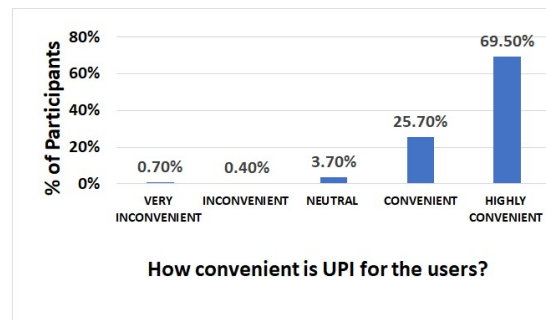


Figure 7: Convenience of making payments using UPI

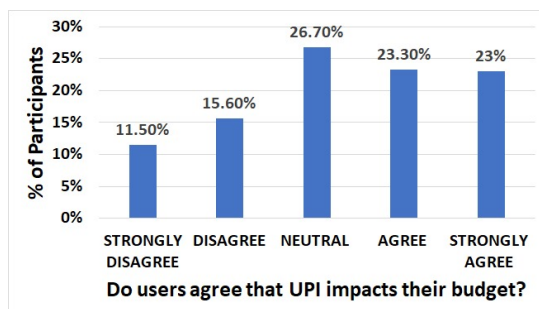


Figure 4: UPI Impact on Budget

paisa (0.0022 USD) decline in debit card transactions, leading to a decrease in ATM visits from 16 to 8 times a year [44]. Additionally,

UPI transactions create a digital record that helps users maintain a record of transactions and financial documentation, collectively contributing to high satisfaction levels among UPI users.

“I love UPI! It’s a fantastic way to spend money without dealing with change or carrying cash. You can pay just Rs 1 (0.012 USD) to a whopping 1 lakh rupees (12,028.79 USD) in one go. It’s one of the best things in this progressive era.”-[P1]

Certain respondents, like a lawyer and clinic owner, emphasized UPI’s role in expediting payments and enhancing business efficiency and financial processes. Notably, UPI’s absence of processing fees for peer-to-peer transactions stands out, benefiting merchants and customers and distinguishing it from credit and debit cards.

“I encourage my customers to use UPI or cash for payments instead of credit or debit cards to avoid the 2-3% transaction

fees, which either come from my earnings or are passed on to the customer, potentially harming my reputation. UPI offers a real-time, no transaction fees and transparent payment solution.”-[P2]

The rise of UPI adoption brings risks, with a surge in scams and fraud. Respondents shared phishing incidents, where malicious actors posed as known contacts or UPI platforms to trick users into sharing sensitive details raising concerns about user vulnerability to cyber threats. Notably, in 2022-23, over 95,000 fraud cases related to UPI transactions were recorded, a notable increase from 84,000 incidents in 2021-22 [47]. To address these issues, users can take protective steps like avoiding sharing sensitive information like OTPs and bank details, updating UPI apps, enabling two-factor authentication, and being cautious with unsolicited payment requests.

“I once got a message asking me to send money, and they wanted my bank OTP. Sadly, I gave it to them, and things got messed up. They took a big chunk of cash, around Rs 10,000, from my bank account. Fortunately, my father worked at a bank and helped fix it. I learned a harsh lesson about being careful with UPI and not sharing important info.”-[P3]

4.2.2 FACTORS INFLUENCING SPENDING BEHAVIOR.

A few respondents strongly preferred using UPI as a payment option as it simplifies transactions, reduces dependency on cash, and encourages frequent expenditure. This shift contributes to a more efficient, cashless economy, revolutionizing how people handle their finances and everyday purchases.

“I spend more often with UPI because you pay directly from your bank account. UPI combined with the rise of Swiggy or Zomato or Zepto or Blinkit, etc., ordering online has become much more appealing and easy. Without UPI, I might not have been spending much on food online. Sometimes, I regret purchases because you only need a phone with Wifi/mobile data”-[P4 & P5]

This digital payment method has reshaped how individuals perceive and manage their finances, sometimes leading to more frequent and impulsive spending patterns. Many individuals struggle with financial regret from overspending, the need for expense mindfulness, and less-than-ideal financial choices. The root cause lies in the stark contrast between physical cash and the abstract nature of digital transactions like UPI payments. Physical cash provides a tangible connection, whereas digital transactions involve mere numbers on a screen, leading to spending without much thought and subsequent feelings of regret and guilt.

“Digital money does not give you guilt when spending. When you spend paper money, you feel guilty. Cash feels tangible, like giving away something valuable when we spend it. But with UPI, it’s just a number on the screen, and it doesn’t feel the same way.”-[P6 & P7]

A diverse range of insights emerged when we asked users about their spending habits and where they generally used UPI. It was evident that UPI has found its way into various aspects of daily life, transforming how people manage their expenses. Users reported using UPI for essentials like rent, groceries, and bills, as well as for work-related expenses, leisure activities such as dining out and entertainment, education, online shopping, and even small everyday indulgences like street food and auto-rickshaw fares.

UPI’s integration into numerous applications has positioned it as a key player in personal and professional finances, particularly in critical business transactions. Its omnipresence showcases its transformative impact on the Indian economic landscape. Respondents acknowledge UPI’s significant influence on spending behavior, budgeting habits, and financial discipline. The study identifies distinct and common patterns in UPI adoption among various user groups, emphasizing its broad utility in routine transactions and reshaping spending habits.

4.2.3 ATTITUDE TOWARDS FINANCIAL MANAGEMENT.

Respondents highlight the significance of financial proficiency in UPI usage, stressing the need for personal finance education and expenditure restraint. Attitudes toward financial discipline varied, reflecting different comfort levels in managing finances using UPI, indicating individual variability in its impact on personal finances.

“Personally, there are occasions when I exceed my initial spending plans, but I rely on finance tracking applications to ensure I stay within my budget.”-[P8]

4.2.4 RECOMMENDATIONS FOR FINANCE TRACKING APPLICATIONS.

Users expressed the need for visualization tools like graphical representations and pie charts to help them understand their spending patterns. The ability to track expenses over time and categorize them was also a common request. Users want insights into where their money is going, allowing them to make more informed financial decisions.

“I’d like my UPI app to allow me to restrict spending in certain categories, such as food, by using QR codes to track locations and deny transactions in those places. I want the app to generate a pie chart showing the average payments I make in different categories over a specific period.”-[P9]

Responses from various professional backgrounds revealed that the specific needs of their professions often shape users’ expectations of financial management apps. Business professionals sought tools to manage business-related finances efficiently, while doctors aimed to collect patient payments and related transactions effectively. These insights underscore the importance of tailoring financial apps to meet the unique requirements of diverse professions, ensuring effective financial management across various occupational backgrounds.

“As a doctor, it would be beneficial if I could easily keep track of all the patients who have paid their fees. Checking transaction histories repeatedly can be quite a hassle. So, having an automatically generated list of patients who have made payments, with a clear categorization of whether the payment has been completed, would be incredibly convenient.”-[P10]

Some respondents went beyond traditional budgeting and desired financial education and guidance within the app, along with receiving notifications and alerts when approaching spending limits or exceeding budgeted amounts.

“Besides handling expenses, it would be great if the app also offers guidance on achieving financial stability and discipline by providing tips and educational content to help users make better financial decisions.”-[P11]

5 DISCUSSION

Answering **RQ1** as mentioned in Section 1, the study results reveal a significant impact of UPI on Indian users' spending habits. Most respondents (74.2%) accepted that UPI adoption led to increased spending, highlighting the UPI influence on user behavior. The data also indicates that many respondents started using UPI during significant economic and social changes, such as demonetization and pandemic-related lockdowns. Furthermore, 59.8% of respondents acknowledged exceeding their budget or overspending due to UPI usage. The qualitative analysis supports these findings, indicating that UPI has contributed to increased spending for most users. The abstract nature of digital transactions leads to psychological shifts, contributing to impulsive spending. In some cases, regret and guilt hit people after they realize their spending by checking their bank balances or statements, compared to spending physical cash, highlighting the lack of tangibility associated with UPI payments.

Answering **RQ2** as mentioned in Section 1, user satisfaction levels were high, with 91.5% of the survey respondents reporting satisfaction with UPI usage and 95.2% of the survey respondents found making payments via UPI convenient. The qualitative analysis underscores the convenience of UPI transactions, with users praising its speed, efficiency, and the elimination of the need for physical cash. The absence of processing fees, as opposed to credit and debit cards, is also highlighted as a positive factor contributing to user satisfaction. The research findings indicate that users perceive UPI as a secure and efficient payment method, eliminating concerns about carrying large sums of money. This positive perception of convenience and efficiency likely contributes to the widespread adoption and satisfaction with UPI, ultimately shaping users' preferences for cashless transactions and influencing spending behavior.

Answering **RQ3** as mentioned in Section 1, the study identifies variations in UPI adoption across demographic groups, particularly in professions. Working professionals and business owners integrate UPI into financial routines seamlessly, while students use it for academic needs. Despite variations, common spending trends are observed across age groups and professions, showcasing UPI's adaptability to diverse user requirements.

5.1 RECOMMENDATIONS AND DESIRED FEATURES IN UPI & FINANCE TRACKING APPLICATIONS

Our research insights highlight key recommendations for UPI platforms and financial app developers, aiming to elevate the digital payment experience. Users increasingly seek budgeting and expenditure limitation features within UPI apps, enabling them to set spending thresholds for specific categories and receive timely reminders to stay within financial goals. Visualization tools like graphical representations and pie charts should be integrated to enhance user comprehension, offering a clear overview of spending patterns over time.

In addition, there is a growing demand for educational content on responsible financial management embedded within the apps to empower users with financial literacy. Recognizing the diverse needs of different professions is crucial; for instance, business professionals may require tools to manage client payments and categorize

expenses efficiently. Tailoring UPI solutions to cater to various user segments ensures accessibility and relevance.

5.2 FINANCIAL APP ANALYSIS IN INDIA

Our comprehensive analysis of various financial tracking and budget management apps in the Indian market is summarized in Table 1. These applications offer diverse features tailored to meet individual financial requirements, ranging from automated organization of transactional data to robust expense-tracking capabilities. Users can leverage these tools to elevate their financial management skills. However, some notable gaps exist in meeting the recommendations outlined in Section 5.1. Specific features like detailed graphical representations of expenditure patterns and spending thresholds with reminders are unevenly available across platforms. Providing educational content on responsible financial management also varies, with certain apps lacking sufficient resources. Addressing these gaps requires finer control mechanisms for spending limits, real-time notifications, goal-based savings integration, AI-powered financial insights, customizable reporting, collaborative budgeting, integration with financial education platforms, and improvements in expense categorization. Enhancing user convenience should involve autonomously pulling expense data without manual input, ensuring a seamless experience while safeguarding user privacy. Balancing automation with privacy protection is crucial for advancing a comprehensive and empowering financial tracking solution for users.

In conclusion, while existing financial tracking apps offer valuable features, there is room for improvement in aligning with the recommendations outlined. Future developments should focus on bridging these gaps to provide users with a more holistic and empowering financial management experience.

Stakeholders, including app developers, government agencies, and users, stand to benefit from incorporating our suggestions into future financial tracking app developments. This comprehensive analysis addresses user needs and provides valuable insights for enhancing the financial management ecosystem, ultimately advancing effective and user-centric solutions.

5.3 FINANCIAL DISCIPLINE AND UPI USAGE

Our research highlights the importance of financial discipline in UPI usage. Users with strong financial discipline leverage UPI for convenience and stability, while those prone to overspending often cite a lack of financial literacy. The abstract nature of UPI transactions can lead to impulsive purchases. Promoting financial education is essential to address this, empowering users to understand expenses and practice responsible spending while embracing UPI. This dual approach empowers users to benefit from UPI while safeguarding their financial well-being.

5.4 FUTURE IMPLICATIONS AND POLICY CONSIDERATIONS

The research outcomes have significant implications for stakeholders in India. Policymakers can design regulations promoting responsible UPI usage and protecting users from scams. Businesses and financial institutions should tailor services to diverse user needs, enhancing overall user experience. Educational initiatives can focus

	FEATURES	YEARLY SUBSCRIPTION COST	PROS	CONS	SUPPORTING PLATFORMS
FinArt [24]	Automatically organizes transactional SMS into expenses, budget, and bill reminders, even when the app is closed.	Rs 799 (9.61 USD)	Data stored on Drive, Auto Tracking, spend analysis dashboards, and budget setting. Separates personal and business accounts for comprehensive financial management.	Setup complexity, ads persist after subscription, and occasional difficulty in SMS tracking.	ANDROID, PC, IOS
Money Manager [33]	The app efficiently handles dual accounts, ensuring separate tracking of income and expenses. It offers offline/cloud backup and Data Export functionalities.	Free	Diverse payment method profiles, encompassing cash and cards, support competitive analysis. Easily create new profiles based on specific needs.	The application takes a long time if currencies are changed to optimize expenses and income. Most fees are non-editable.	ANDROID, IOS
Monify [25]	The app is straightforward; the infographics make it vibrant and easy to use.	Rs 180 (2.17 USD)	Set bill reminders, manage multiple accounts and easily import data through Excel.	Frequent prompts for premium upgrades	ANDROID, PC, IOS
Wallet [20]	Users can use it for different currency rates worldwide. The Group Sharing option is also available in the application	Rs 700 (8.42 USD)	Monitor group expenses in the totals tab, track recent costs in the activity tab, and official API support available.	Accounts auto-disconnect; unable to modify the initial login currency.	ANDROID, IOS
Splitwise [10]	App simplifies bill splitting, not for personal expense tracking. Form groups, manage shared bills, and track dues effortlessly.	Rs 499 (6 USD)	Clean UI. Manages multiple wallets, links bank accounts or Crypto wallets directly, and tracks transactions seamlessly (premium feature).	Tracking and analyzing personal finances can be challenging.	ANDROID, PC, IOS
Spendee [18]	The app provides user-friendly expense and income tracking with easy-to-use tags for sorting costs efficiently.	Rs 900 (10.43 USD)	Effortlessly track expenses, share budgets, view spending analytics, and easily monitor debt progress.	Certain features are restricted behind a paywall.	ANDROID, PC, IOS
GoodBudget [6]	Budget management application. The app lets you create and refill multiple budgets (envelopes) at intervals like monthly or weekly.	Rs 5817 (70 USD)	Create multiple wallets, set up recurring payments, track credit card and debt. Link bank accounts and enable group sharing (premium only).	Simple UI. Adding transactions is tedious, and there's no API support.	ANDROID, PC, IOS

Table 1: Comparison of Personal Finance Tracking Apps

on improving financial literacy and responsible spending habits among UPI users. This underscores the transformative power of UPI, urging all parties to adapt for the benefit of Indian consumers and the broader economy. Many other countries, like France, Singapore, etc., have started embracing UPI to promote digital payments [43]. With the recent developments by NPCI like UPI integration with Rupay Credit Card [14], tap and pay feature with Near Field Communication (NFC) [3], UPI Lite [40] and UPI LiteX [44] for offline payments, the infrastructure of UPI is being aimed to scale to 100 billion monthly transactions, making it a greater force of revolution in the Indian financial landscape.

6 CONCLUSION

In this paper, we conducted a user study to understand the influence of UPI on spending behavior among Indian users. We studied how UPI has enabled convenience and seamless financial transactions, overshadowing cash transactions. UPI has played a pivotal and remarkable role in India's push for a cashless economy and is also benefiting India's GDP. Our study culminates by aligning findings with research questions and user recommendations, providing actionable insights for enhancing UPI app features and facilitating effective expenditure management. This research has broader implications for policymakers, businesses, and financial institutions, highlighting the need to adapt and promote responsible UPI usage to maximize its transformative potential in the Indian economic landscape.

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A SURVEY QUESTIONS

1. What is your age?
2. What is your profession?
3. What is your gender? (Options: Male, Female, Non-binary, Prefer not to say)
4. Have you used UPI as the payment method?
5. If yes, which year did you start using it?
6. How often do you use UPI for making payments? (Daily, Weekly, Monthly, Occasionally, Rarely)
7. Have you noticed any changes in your spending habits since you started using UPI?
8. On a scale of 1 to 5, rate the convenience you find with UPI for making payments.
9. How satisfied are you with your overall experience using UPI for payments? (1 = Not Satisfied at all, 5 = Extremely Satisfied)
10. On average, how much do you spend daily using UPI?
11. Have you found any change in your spending ever since using UPI?
12. Do you feel that using UPI affects your ability to stick to your budget?
13. Have you experienced any instances where you have exceeded your budget or spent more than intended due to UPI usage?
14. How often do you check your transaction history or keep track of your UPI spending? (options: rarely, occasionally, frequently, very frequently, regularly)
15. How do you manage/track your expenses? (Which app do you use, if you do, e.g., split-wise, etc)
16. Additional Comments: Do you want to share anything about your experience with UPI and its impact on your spending behavior?
17. Will you be available for online interviews if called? (Yes/No)
18. If yes, provide your email address :
19. If yes, provide your phone number :

B INTERVIEW QUESTIONS

Demographic Information:

- a. What is your age?
- b. What is your gender? (Options: Male, Female, Non-binary, Prefer not to say)
- c. What is your occupation?

Usage of UPI:

- a. Have you used UPI as a payment method? (Yes/No)
- b. If yes, when did you start using it?
- c. How often do you use UPI to make payments? (Daily, Weekly, Monthly, Occasionally, Rarely)
- d. Can you briefly describe your experience with using UPI to make transactions?

- e. Have you noticed any changes in your spending habits since you started using UPI? (options: "Increased spending," "Decreased spending," or "No change")

- f. If yes, could you elaborate on the changes you have observed in your spending habits after adopting UPI?

Perception of UPI Convenience:

- a. On a scale of 1 to 5, rate the convenience you find with UPI for making payments. (1 = Not Convenient at all, 5 = Extremely Convenient)
- b. In what ways do you perceive UPI to be convenient/inconvenient for you?
- c. On a scale of 1 to 5, how would you rate the convenience of using UPI compared to traditional payment methods like cash? (1 = Not Convenient at all, 5 = Extremely Convenient)
- d. How satisfied are you with your overall experience of using UPI for payments? (1 = Not Satisfied at all, 5 = Extremely Satisfied)

Income Information:

- a. What is your approximate annual income range?

Spending Behavior:

- a. On average, what is the transaction amount when you use UPI?
- b. On average, how much do you spend daily using UPI?
- c. On average, how much do you spend monthly using UPI?
- c1. Where do you spend your money?
- c2. Does the spending amount include only UPI transactions or all spending in general?
- d. Have you found any change in your spending ever since using UPI?
- e. If yes, could you specify the reasons for the increased spending after adopting UPI?
- f. Do you feel that using UPI affects your ability to stick to your budget? (Options: Strongly Disagree, Disagree, Neutral, Agree, Strongly Agree)
- g. How often do you check your transaction history or keep track of your UPI spending? (Options: Rarely, Occasionally, Frequently, Very Frequently, Regularly)
- h. Have you experienced any instances where you have exceeded your budget or spent more than intended due to UPI usage?
- i. In your opinion, does using UPI encourage you to spend more money overall? (Options: Strongly Disagree, Disagree, Neutral, Agree, Strongly Agree)

Factors Influencing Spending Behavior:

- a. Besides UPI usage, do other factors have influenced your spending behavior?
- b. If yes, please specify those factors.
- c. Are you aware of any psychological factors influencing your spending behavior when using UPI? If yes, please elaborate.

Attitudes Towards Financial Management:

- a. How would you describe your level of financial discipline? (1 to 5)
- b. Do you feel that using UPI has made managing your finances easier or more difficult? ("Easier," "More Difficult," "No Change")
- c. How do you rate the statement "Do you feel that UPI provides you with a clear understanding of your spending patterns"? (Options: Strongly Disagree, Disagree, Neutral, Agree, Strongly Agree)

d. How do you rate the statement "UPI has simplified your financial management"? (Options: Strongly Disagree, Disagree, Neutral, Agree, Strongly Agree)

Additional Comments:

a. Is there anything else you would like to share about your experience with UPI and its impact on your spending behavior?

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